

MP UNIT 2

Topics

- **80386DX Signals-** Pin Diagram of 80386 and description.
- **80386DX Bus Cycles-** System Clock, Bus States, Pipelined and Non-pipelined Bus Cycles.
- **80387 NDP-** Numeric data processor

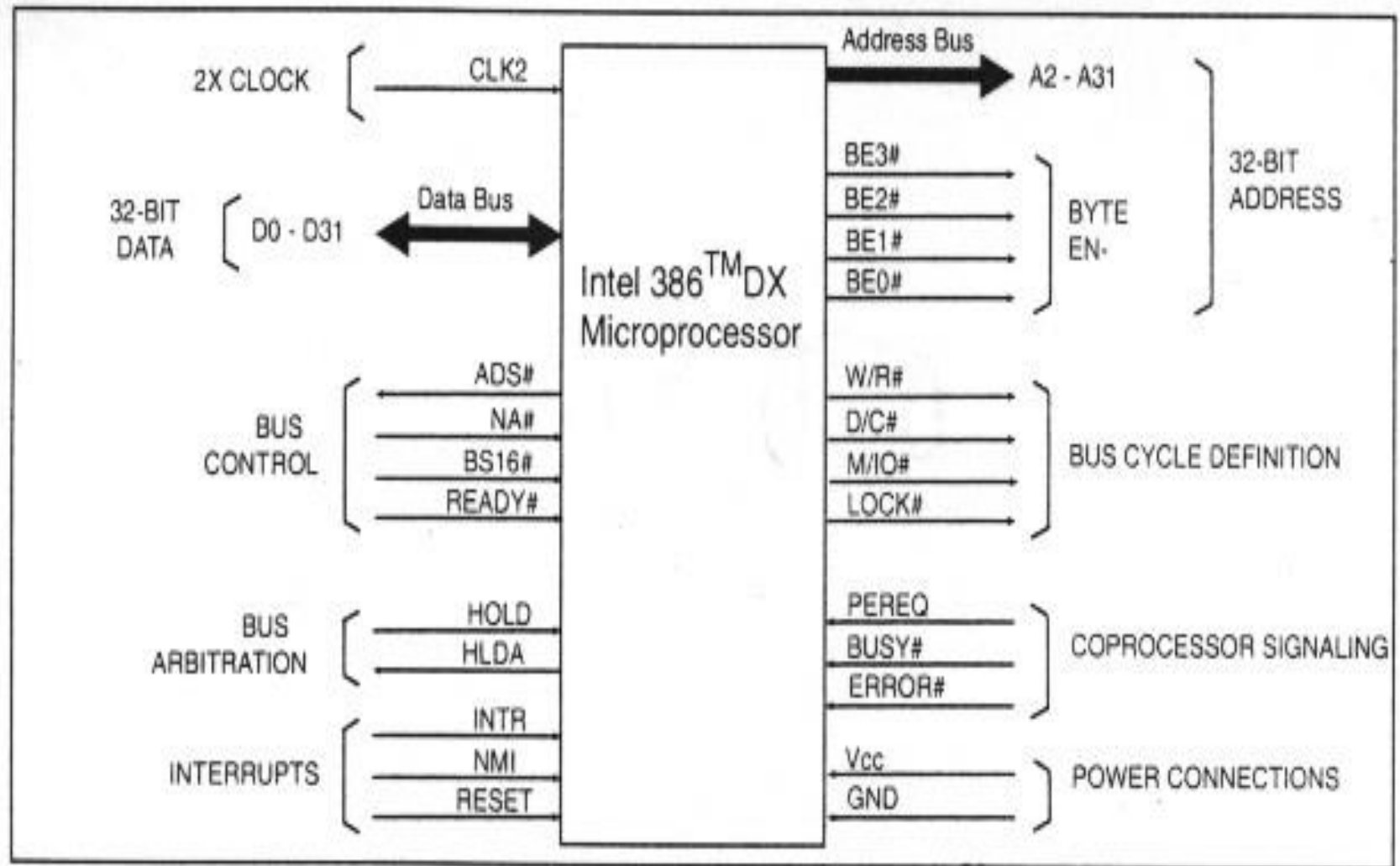
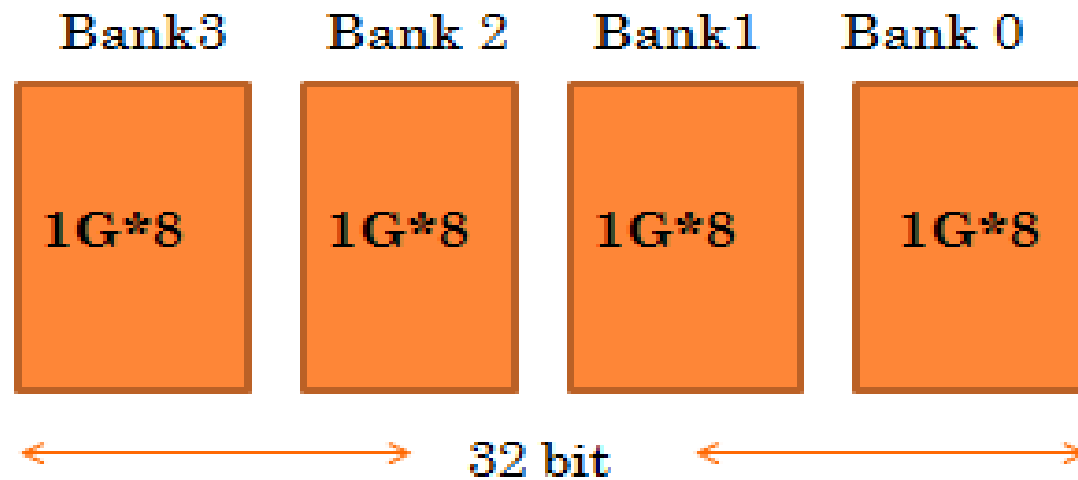


Figure 13-2. 80386 Block Diagram (# indicates active low)

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- **CLK2**: The input pin provides the basic system clock timing for the operation of 80386.
- **D0 – D31**: These 32 lines act as bidirectional data bus during different access cycles.
- **A31 – A2**: These are upper 30 bit of the 32- bit address bus.
- **BE0 to BE3**: (Active Low) The 32- bit data bus supported by 80386 and the memory system of 80386 can be viewed as a 4-byte wide memory access mechanism.



BE0, BE1, BE2 and BE3 are active low signals.

Bus Control

ADS#: (Address Data Strobe)

Active when **issued a valid request.**

The address status output pin indicates that the address bus and **Bus Cycle Definition Pins**(W/R#, D/C#, M/IO#, BE0# to BE3#) are carrying the respective valid signals.

BS16#: The bus size – 16 input pin **allows the interfacing of 16 bit devices with the 32 bit wide 80386 data bus.**

- **READY#**: The ready signal indicates to the CPU that the **previous bus cycle has been terminated** and **the bus is ready for the next cycle**. The signal is used to insert WAIT states in a bus cycle and is useful for interfacing of slow devices with CPU.
- **NA#**: Gives address of next instruction if pipelining is enabled. If pipelining is not enabled, this pin is high, if instruction is in waiting state.

Bus Arbitration

- **HOLD**: The Bus hold input pin enables the other bus masters to gain control of the system bus if it is asserted.
- **HLDA**: The bus hold acknowledge output indicates that a valid bus hold request has been received and the bus has been relinquished by the CPU.

Interrupts

- **INTR**: This interrupt pin is a mask-able interrupt, that can be masked using the IF of the flag register.
- **NMI**: A valid request signal at the non-mask-able interrupt request input pin internally generates a non-mask-able interrupt of type 2.
- **RESET**: A high at this input pin suspends the current operation and restart the execution from the starting location.

Co-Processor Signaling

- **BUSY#**: The busy input signal indicates to the CPU that the **coprocessor is busy** with the allocated task.
- **ERROR#**: The error input pin indicates to the CPU that the coprocessor has encountered an error while executing its instruction.
- **PEREQ: (Processor extension request)**
Output signal indicates to the **CPU to fetch data**.

A bus cycle definition pins

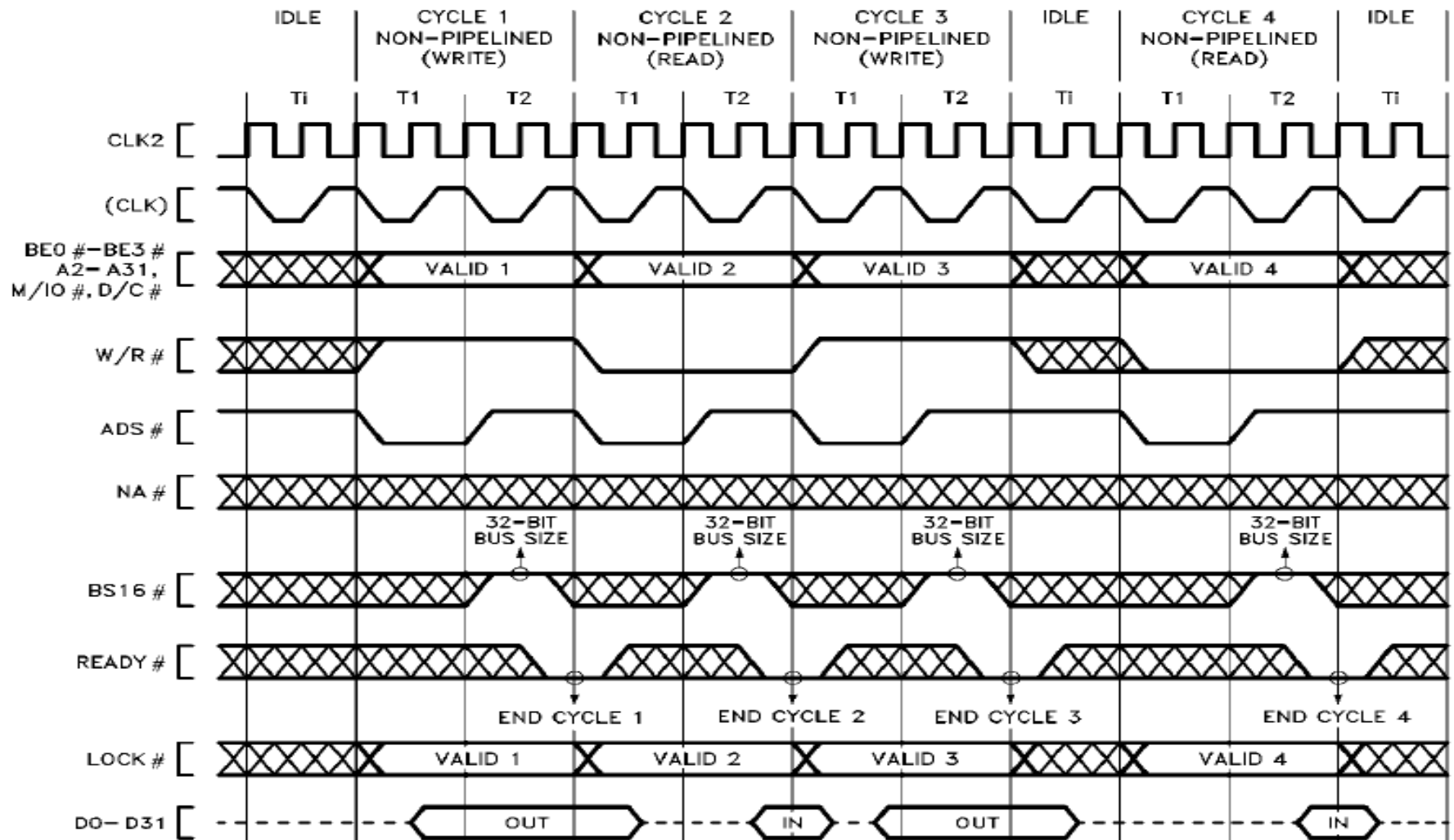
- **LOCK#**: BUS LOCK is a bus cycle definition pin that indicates that system have locked system bus.
- **W/R#**: WRITE/READ is a bus cycle definition pin that distinguishes write cycles from read cycles.

- **D/C#**: DATA/CONTROL is a bus cycle definition pin that distinguishes data cycles, either memory or I/O, from control cycles which are: interrupt acknowledge, halt, and instruction fetching.
- **M/IO#**: MEMORY I/O is a bus cycle definition pin that distinguishes memory cycles from input/output cycles.

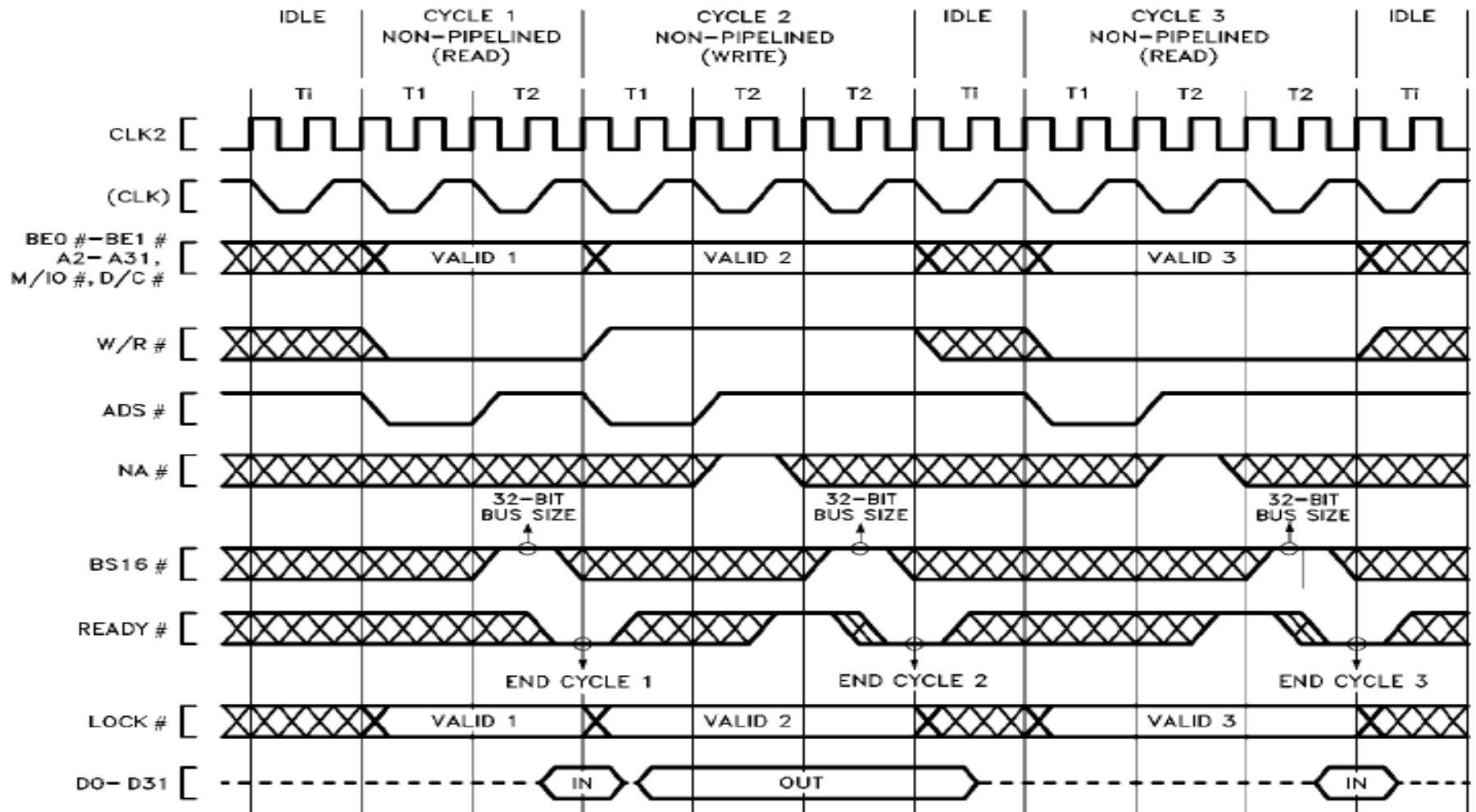
Power Connection Pins

- **VCC:** These are system power supply lines.
- **GND:**

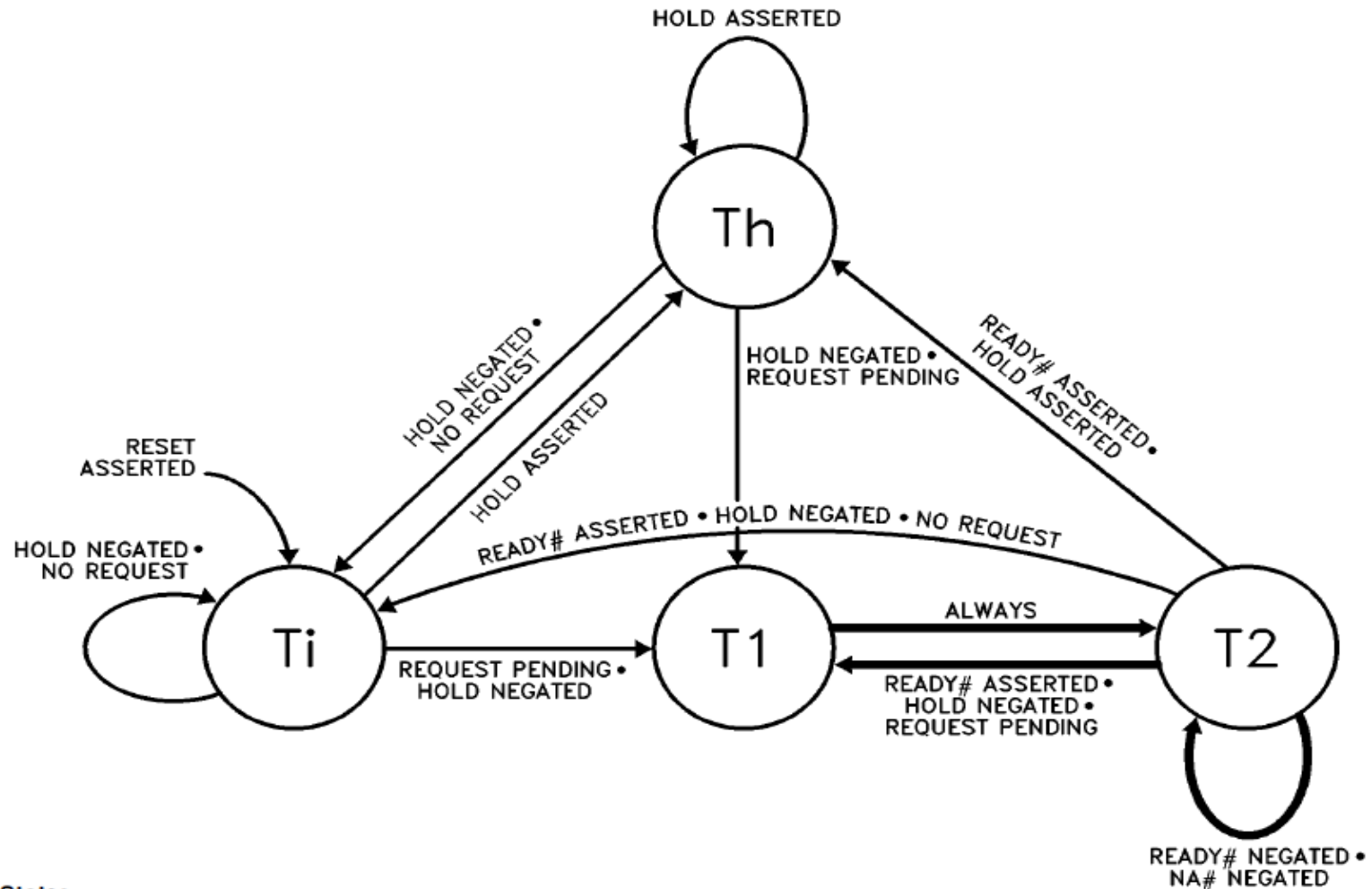
Non-pipelined read & write cycles (No wait states)



Non-pipelined read & write cycles (With wait states)

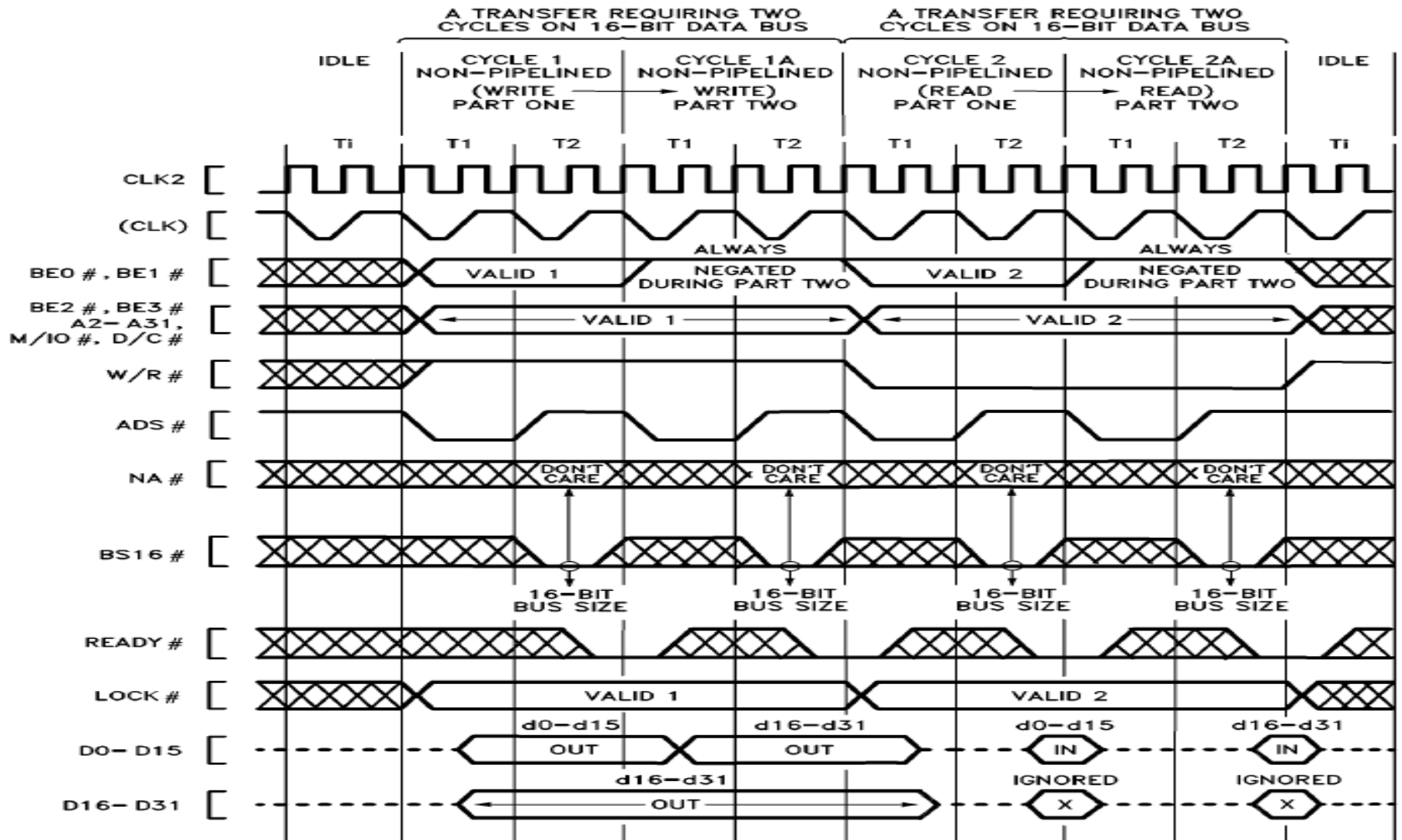


Bus States (Using non-pipelined address)

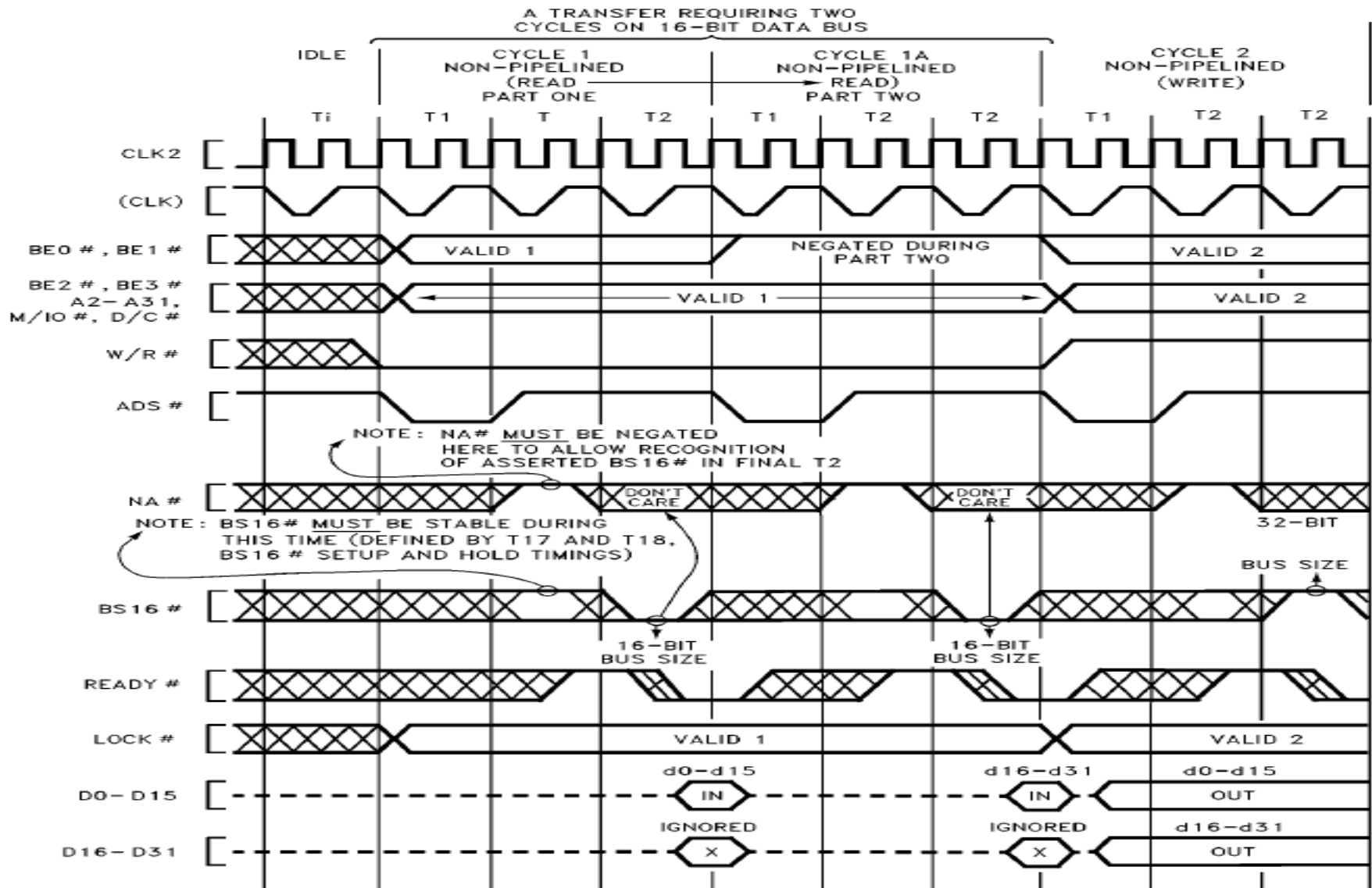


- **T1:** First clock of a non-pipelined bus cycle (Intel386 DX drives new address and asserts ADS#)
- **T2:** subsequent clocks of a bus cycle when NA# has not been sampled asserted in the current bus cycle
- **Ti:** idle state
- **Th:** Hold acknowledge state (Intel386 DX asserts HLDA)
- The fastest bus cycle consists of two states: T1 and T2.

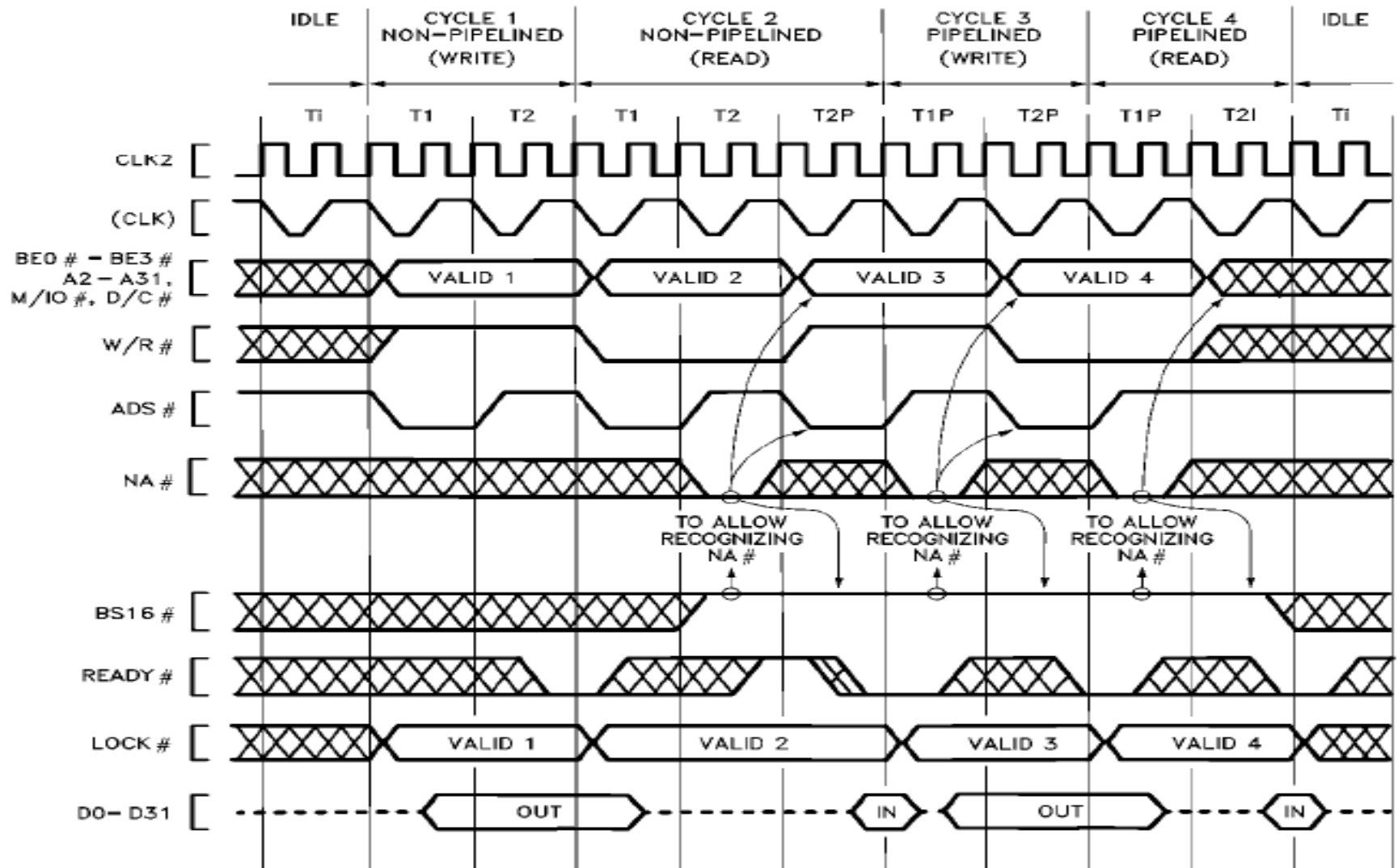
Asserting BS16#: No wait states



Asserting BS16#: Wait states



Transitioning to pipelined address



Fast Transitioning to pipelined address

